

Tuan Ehshan Pallie

Textile and Apparel Engineering undergraduate (B.Sc. Hons, University of Moratuwa) with a strong foundation in fibre science, fabric and weave structures, fibre orientation and material testing. Experienced in parametric CAD (Fusion, Inventor), endurance analysis and prototyping of structured textile substrate. Disciplined team player with leadership and coaching experience.

Contact

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https://tinyurl.com/2w5prcv7

Colombo, Sri Lanka

Language

English *(Fluent)*
Sinhala *(Native)*
Tamil *(Fluent)*
Urdu *(Basic)*

Skills

- Languages: C#, C++, Java, Python, JavaScript, PHP
- Hardware & IoT: microcontrollers, ESP32, Arduino Uno, Bio Sensors
- CAD & Tools: Fusion, Inventor (parametric modelling, endurance analysis), GIT, MS Office
- Materials: fibre science, weave architectures, fibre orientation, stiffness & strength
- Teamwork and Collaboration
- Material testing & characterisation
- Composites basics: carbon/glass fibre, epoxy resin, layup & curing
- Research Skills
- Hands-on workshop work, safety & process discipline
- Excellent Communication Skills

Experience

Assistant Chess Coach *Nov 2024 – Jan 2026*

[Royal Institute International School, Nugegoda, Maharagama, Havelock]

- Helped Students gain and improve ratings (300+ students)
- Assisted students in achieving gold, silver and bronze medals in national, international events.

National Trainer - Certified by Trainers Commission, Chess Federation of Sri Lanka

Education

B.Sc. (Hons) Textile and Apparel Engineering *Jan-2025*

University of Moratuwa | Fibre Science, Fabric Mechanics, Textile Testing

Bachelor of Information Technology *June-2025*

University of Moratuwa

G.C.E. A/L – Physical Science Stream *2023*

Royal College, Colombo 05

Combined Mathematics - A Physics - A Chemistry - B

G.C.E O/L *2019*

Isipathana College, Colombo 05

9A passes

Projects & Research

Undergraduate Research – University of Moratuwa

Mitigating chemical pollution from textile wastewater: A sustainable engineering approach for water protection and public health in Sri Lanka | 2026

- Evaluated material interactions and treatment methods for textile wastewater.
- Abstract accepted - International Conference on Engineering for Humanity - ICEH 2026 (Pending Publication)

Patient Monitoring Glove

- Developing a wearable textile, based patient monitoring system using an Esp32 microcontroller
- Integrating PPG, temperature and biosensing technologies for continuous physiological monitoring.
- Engineered an elastomeric textile substrate (20% spandex blend) for mechanical recovery, tension stability and sensor retention.
- Modelled gloves in Fusion/Inventor; analysed endurance and fit under cyclic loading.
- Developing alert mechanisms for abnormal heart rate and glucose conditions

HealthyByte (BMI & Blood Pressure Monitoring System)

- Calculates BMI levels and Blood Pressure levels according to medical standards
- 50+ patient telemetry included

Sports & Achievements

- Asian Youth Chess Championship, Sri Lanka - Silver Medal
- World Schools Chess Championship, Turkey - Top 10
- Commonwealth Chess Championship, India - Participated
- National Schools Individual Chess Championship - Champion
- National Youth Individual Chess Championship - 1st Runners up
- National Schools Team Chess Championship - Champions
- Sri Lanka University Games Champions - University of Moratuwa
- University of Moratuwa, Western Province, Royal College Colorsman.
- Yellow Belt in Taekwondo

Leadership

- President - Entrepreneurship Society, University of Moratuwa
- Vice President - Leo Club of Polgasowita - 2026/27